

Lecture 5: Causal Inference - Part 1

LSE ME314: Introduction to Data Science and Machine Learning (<https://github.com/me314-lse>)

2026-07-20

Daniel de Kadt

Where Did We Begin?



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Imagine you're a member of Netflix's c-suite. What kind of questions might you have about Squid Game?

- Did viewers like Squid Game?
- Which viewers most liked Squid Game?
- What types of viewers engaged with Squid Game?
- Did Squid Game increase Netflix viewership?
- What was the \$ value of Squid Game?
- Who should we advertise or push Squid Game to?
- How many \$ should we spend on advertising Squid Game?
- Should we invest in Season 4?
- What can we learn from Squid Game about the types of shows that are successful?

What Good is a Regression?

To answer these questions, we spent last week building up to the idea of the conditional expectation function (CEF) of Y :

$$\mathbb{E}[Y|X] = f(X)$$

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 - Bivariate: Comparing values of Y for different values of X
 - Multivariate: Comparing values of Y for different combinations of features

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- Why do we care? Because many of those previous questions are answered via comparison:
 - Bivariate: Comparing values of Y for different values of X
 - Multivariate: Comparing values of Y for different combinations of features
- We can use regression (bivariate or multivariate) to make these comparisons

What Good is a Regression?

Recall that we can use regression in at least three ways:

1. As a **causal device**:

- Estimate the effect of a feature on the outcome
- Comparison: if X changes, does this *change* Y ?
- E.g. 'Did Squid Game increase Netflix viewership?'

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 - Summarize co-occurrence of two variables, irrespective of DGP
 - Comparison: for different levels of X , is Y usually different?
 - E.g. 'Which viewers most liked Squid Game?'
3. As a **predictive device**:
 - Fill in 'missing values' (unseen realizations) of Y using X
 - Comparison: if we saw a new value of X , what would we expect Y to be?
 - E.g. 'Who should we advertise or push Squid Game to?'

Goals for Today

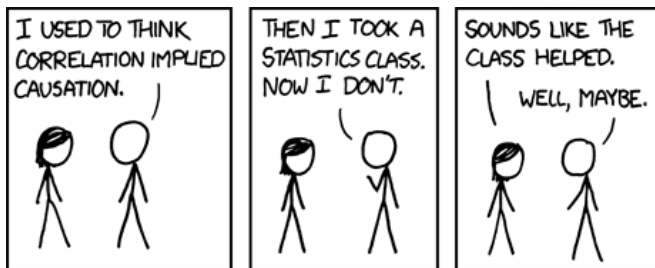
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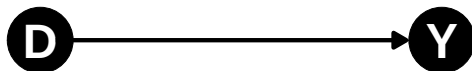
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Theories of Causation

Cause and Effect



D: Our 'causal feature/variable,' often called a 'treatment'

Y: Our 'outcome variable', or 'response variable'

Cause and Effect

Before we can get to the good stuff, we have to do a little theory:

1. Define causes and effects in abstract terms
2. Build two formal representations of causation:
 - Potential outcomes
 - Graphical models

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One important implication is that causal variables must be **manipulable**:

*No **causation** without **manipulation**.* – Holland, 1986

Good Causal Questions

Manipulability means we must think very carefully about causal questions. . .

1. (Largely) immutable characteristics:

- Customers' sex assigned at birth → consumer preferences
- Race and ethnicity → employment outcomes
- Country of origin → ideology

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1. (Largely) immutable characteristics:

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2. Non-successive chains:

- Monthly expenditure → monthly savings
- Platform decision made in 2012 → customer behaviour today

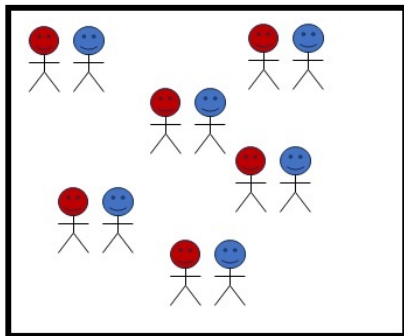
Potential Outcomes

We have six participants in a study with a binary treatment:



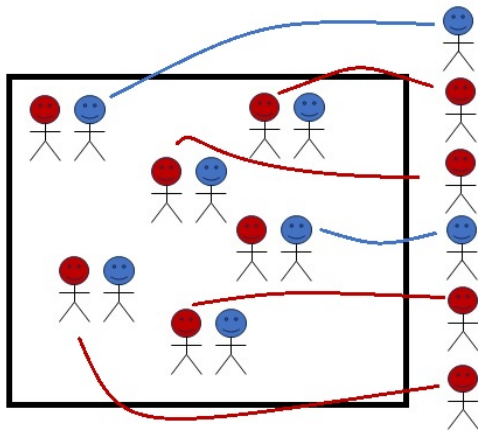
Potential Outcomes

Each has **two** potential outcomes:



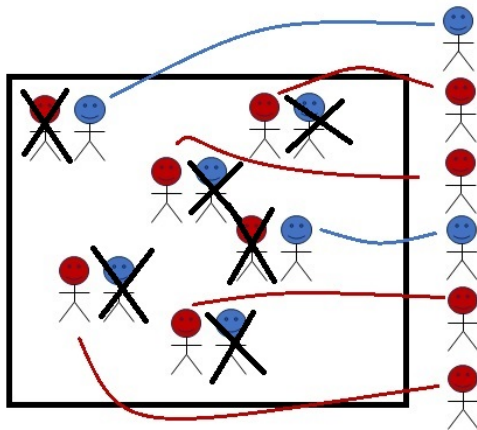
Potential Outcomes

Only **one** is realised, based on treatment status:



Potential Outcomes

The **Fundamental Problem of Causal Inference** (FPCI):



Potential Outcomes

Let's do this more formally, assuming some treatment D on an outcome Y .

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For a binary treatment ($D \in \{0, 1\}$), two potential outcomes:

- $Y_i(0)$: The value Y would take if they did not receive treatment ($D_i = 0$)
- $Y_i(1)$: The value Y would take if they did receive treatment ($D_i = 1$)

These are mutually exclusive **counterfactual** quantities: Only one can ever be realized.

Potential Outcomes

Let's go back to the toy example, but this time in R.

We have six students, each with two potential outcomes:

```
students <- data.frame(  
  student = c("Radha", "Pam", "Konstantinos",  
              "Joy", "Shawn", "Brooke"),  
  Headache_0 = c(9, 4, 8, 3, 10, 3),  
  Headache_1 = c(7, 1, 9, 5, 7, 3)  
)  
  
students
```

	student	Headache_0	Headache_1
1	Radha	9	7
2	Pam	4	1
3	Konstantinos	8	9
4	Joy	3	5
5	Shawn	10	7
6	Brooke	3	3

Potential Outcomes

So, let's distribute or 'assign' a **treatment**:

```
students <- students %>%  
  mutate(Medicine = sample(c(0, 1), nrow(students), replace = TRUE))  
  
students
```

	student	Headache_0	Headache_1	Medicine
1	Radha	9	7	1
2	Pam	4	1	1
3	Konstantinos	8	9	1
4	Joy	3	5	0
5	Shawn	10	7	1
6	Brooke	3	3	1

Potential Outcomes

The **realized outcomes** are a function of potential outcomes and treatment:

```
students <- students %>%  
  mutate(Headache = case_when(  
    Medicine == 0 ~ Headache_0,  
    Medicine == 1 ~ Headache_1,  
    TRUE ~ NA_real_  
  )  
)  
  
students
```

	student	Headache_0	Headache_1	Medicine	Headache
1	Radha	9	7	1	7
2	Pam	4	1	1	1
3	Konstantinos	8	9	1	9
4	Joy	3	5	0	3
5	Shawn	10	7	1	7
6	Brooke	3	3	1	3

Potential Outcomes to Realized Outcomes

Formally, the realized or observed outcomes are:

$$Y_i = Y_i(1) \times D_i + Y_i(0) \times (1 - D_i)$$

Read:

- When treatment is 1, we get back $Y_i(1)$
- When treatment is 0, we get back $Y_i(0)$.

Potential Outcomes: Estimands

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Let's define the **Individual Treatment Effect (ITE)**:

$$ITE_i = Y_i(1) - Y_i(0)$$

Read: For individual i , the effect of treatment is the difference in that individual's potential outcomes.

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When i does not respond to treatment, in other words: $ITE_i = 0$.

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Individual-level effects are very hard to target. (Why?)

Instead let's focus on two group-level estimands:

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Average Treatment Effect (ATE):

$$ATE = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)]$$

Average Treatment Effect on the Treated (ATT):

$$ATT = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1]$$

Potential Outcomes: Estimands

Let's go back to our toy example to see this more clearly:

```
students %>%  
  summarise(ATE = mean(Headache_1) - mean(Headache_0),  
            ATT = mean(Headache_1[Medicine == 1]) -  
                  mean(Headache_0[Medicine == 1]),  
            ATE_hat = mean(Headache[Medicine == 1]) -  
                  mean(Headache[Medicine == 0]))
```

	ATE	ATT	ATE_hat
1	-0.8333333	-1.4	2.4

Graphical Models of Causation

An alternative (but $\sim$ equivalent) way to represent causal relationships is with a **graphical model**.

We call these **Directed Acyclic Graphs** (DAGs).

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For any causal question, we can write down our beliefs about:

1. The data generating process for Y
2. The data generating process for D

Graphical Models of Causation

For our purposes, a graphical causal model has four features:

- **Nodes**: represent features or outcomes (e.g., X , Y)
- **Edges**: represent causal relationships (e.g., $X \rightarrow Y$)
 - Note, the *absence* of edges implies an *absence* of a causal relationship
- **Directed**: Pairs can be said to have a tail and a head (and in chains, ancestors and descendants)
- **Acyclical**: Nodes cannot terminate in themselves (e.g., $X \rightarrow Y \rightarrow X$)

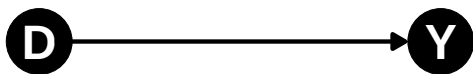
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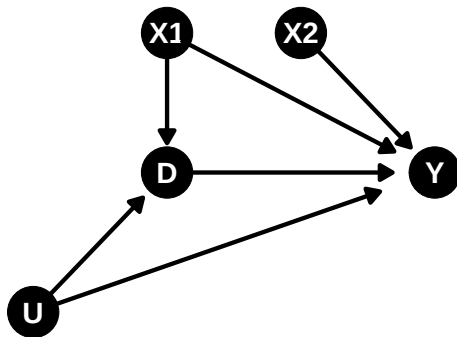
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One key thing to remember: A graphical model is a *theory of the DGPs of Y and D that you assume*.

Graphical Models of Causation



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The Identification Problem

The Identification Problem for Causal Inference

Identification:

In statistics, an **estimand** (parameter) is **identified** if its value can, asymptotically, be uniquely **mapped to** observed data and unidentified otherwise.

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Let's see this problem formally.

Selection Bias

The naïve difference of observed means in the treatment groups:

$$\underbrace{\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0]}_{\text{Observed difference in average outcome measures}} = \mathbb{E}[Y_i(1) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0]$$

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$$= \underbrace{\mathbb{E}[Y_i(1) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 1]}_{\text{ATT}} + \underbrace{\mathbb{E}[Y_i(0) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0]}_{\text{Selection bias}}$$

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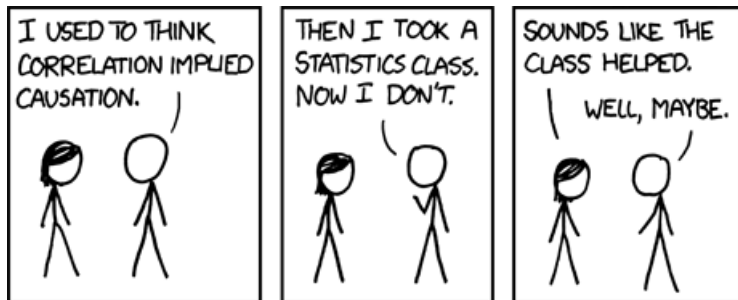
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Correlation (association, observed difference) is not necessarily causation.

Selection Bias

Haha!



Source: Randall Monroe, <https://xkcd.com/552/>

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Canonical example – those who are more risk averse will be more likely to wear a seatbelt:

$$\mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0] > 0$$

The Identification Problem Redux

Identification Strategy:

A combination of **data** and **assumptions** which allows us to **identify** a causal estimand by estimating (“filling in”) the missing potential outcomes (usually at a group level) in expectation.

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Today:

- Randomization
- Selection on Observables (SOO)

Tomorrow:

- Instrumental Variables (IV)
- Regression Discontinuity (RD)
- Difference-in-Differences (DiD)

Randomization

Randomization: Setup

As before, our setting:

- Y : Outcome variable
- $D \in \{0, 1\}$: Treatment variable (binary)
- $Y(0)$: Potential outcome if $D = 0$
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Assume that D is assigned by an understood random **assignment mechanism**

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This is called a 'randomized experiment', 'randomized controlled trial' (RCT), 'A/B test'.

There are many ways to do this (simple, complete, clustered, etc.).

Randomization: Independence

Assumption: $D \perp \{Y(0), Y(1)\}$

Read: Treatment status (D) is independent of potential outcomes.

We will call this assumption **independence** (aka ignorability, randomization)

Intuition: By randomizing treatment, we have severed any relationship between **treatment** and **potential outcomes**.

Check: Does the independence assumption imply $D \perp Y$?

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Check: Does the independence assumption imply $D \perp Y$? No!

Randomization: Identification (POs)

How does independence help us? Recall selection bias:

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If $D \perp \{Y(0), Y(1)\}$, then:

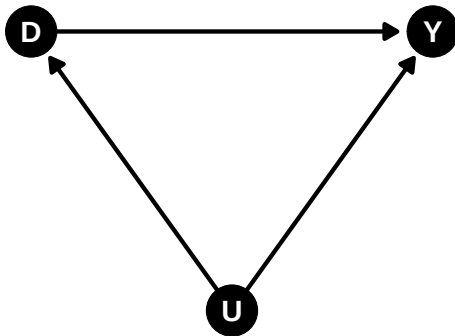
$$\mathbb{E}[Y_i(0) \mid D_i = 1] = \mathbb{E}[Y_i(0) \mid D_i = 0]$$

Thus, the selection bias term is zero and:

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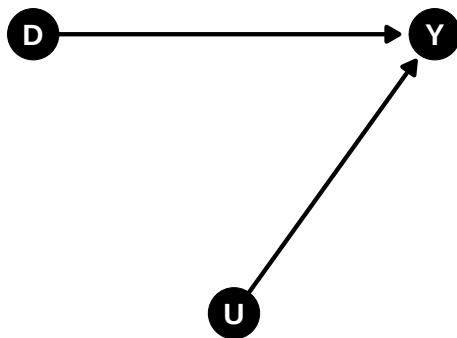
QED

Randomization: Identification (DAGs)



Here U is a canonical **confounder** – it both **selects** individuals into treatment and **affects** the outcome.

Randomization: Identification (DAGs)



Under randomization, the $U \rightarrow D$ is severed – there is no longer confounding.

Randomization: Estimation

In a randomized experiment, assuming independence, we have proven that:

The observed **difference in means** (D-i-M) is an unbiased estimator of the ATT.

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The observed **difference in means** (D-i-M) is an unbiased estimator of the ATT.

It turns out that because D is randomly assigned, it is *also* an unbiased estimator of the ATE:

$$\begin{aligned}\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] &= \mathbb{E}[Y_i(1) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0] \\ &= \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1] \quad (\text{by independence}) \\ &= \mathbb{E}[Y_i(1) - Y_i(0)]\end{aligned}$$

So, in a randomized experiment, you can just calculate the group D-i-M, and use a t -test.

Randomization: Estimation

Let's do just that on a slightly larger dataset than our toy example:

```
t.test(Headache ~ Medicine, data = students_analysis,  
       var.equal = FALSE)
```

Welch Two Sample t-test

data: Headache by Medicine

t = -17.658, df = 918.16, p-value < 2.2e-16

alternative hypothesis: true difference in means between group 0 and group 1 is

95 percent confidence interval:

-3.788941 -3.030952

sample estimates:

mean in group 0 mean in group 1

5.614484

9.024431

Randomization: Estimation

An alternative (but equivalent) estimator is linear regression:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 D_i$$

Under independence, $\hat{\beta}_1$ identifies the ATT and ATE.

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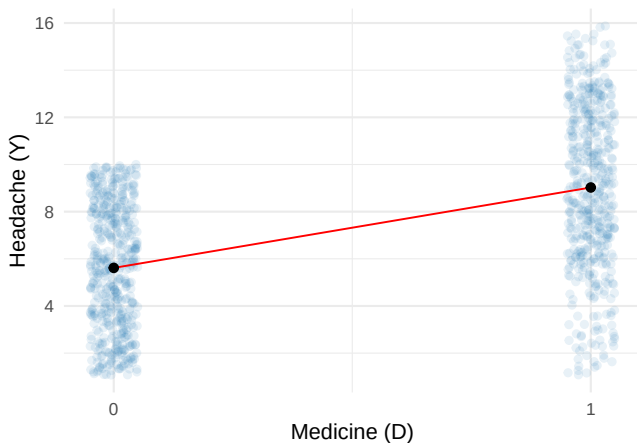
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Remember: Regression is a tool for comparison!

Intuition check: What does $\hat{\beta}_0$ identify? $\mathbb{E}[Y_i(0)]$

Randomization: Estimation

Let's see this visually – the fitted line simply connects the two group means:



So the slope $\hat{\beta}_1$ is exactly the difference in means, $\bar{Y}_{D=1} - \bar{Y}_{D=0}$.

Randomization: Asymptotic Inference

Regression gives us inference for free, but don't forget to use robust standard errors:

```
lm_robust(Headache ~ Medicine, data = students_analysis) %>%  
  summary()
```

Call:

```
lm_robust(formula = Headache ~ Medicine, data = students_analysis)
```

Standard error type: HC2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	CI Lower	CI Upper	DF
(Intercept)	5.614	0.1152	48.76	2.637e-266	5.389	5.840	998
Medicine	3.410	0.1931	17.66	6.222e-61	3.031	3.789	998

Multiple R-squared: 0.2387 , Adjusted R-squared: 0.2379

F-statistic: 311.8 on 1 and 998 DF, p-value: < 2.2e-16

Selection on Observables

SOO: Setup

In a randomized experiment **we** control the assignment mechanism. Often that is not the case.

As before, our setting:

- Y : Outcome variable
- $D \in \{0, 1\}$: Treatment variable (binary)
- $Y(0)$: Potential outcome if $D = 0$
- $Y(1)$: Potential outcome if $D = 1$
- X : A (set of) observable pre-treatment covariate(s)

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D is not randomized, so we have to rely on **observed variation in D** , and make some assumptions about it.

This is called **Selection on Observables** (SOO): We believe the selection process is a function of observed features.

SOO: Conditional Independence

Assumption 1: $D \perp \{Y(0), Y(1)\} \mid X$ for any $x \in \mathcal{X}$

Read: Treatment status (D) is independent of potential outcomes, conditional on X .

We will call this assumption **conditional independence** (aka conditional ignorability).

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We will call this assumption **conditional independence** (aka conditional ignorability).

Intuition: Proposes that within each level of X , there is an 'experiment' such that $D \perp \{Y(0), Y(1)\}$.

SOO: Common Support

Assumption 2: $0 < \Pr(D_i = 1 \mid X_i = x) < 1$ for any $x \in \mathcal{X}$

Read: For any value of X_i , i could have received treatment or control.

Intuition: Proposes that each experiment is 'meaningful'

SOO: Common Support

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Read: For any value of X_i , i could have received treatment or control.

Intuition: Proposes that each experiment is 'meaningful'

We will omit the proof, but here is the intuition:

- The effect of D on Y for each value of X is called a **Conditional Average Treatment Effect (CATE)**
- Each CATE is identified (by conditional independence)
- The ATE (or ATT) is just a weighted average of all CATEs (by common support)
- The ATE is identified as a weighted average of CATEs, where weights $\rightarrow P(X_i = x)$

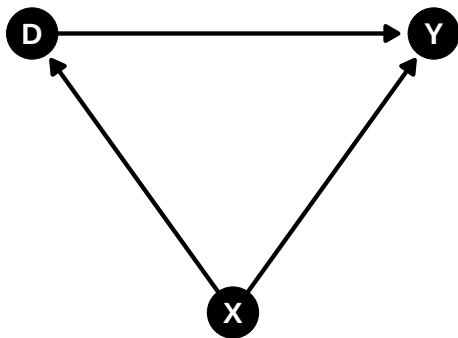
SOO: Identification (DAGs)

We can use DAGs to learn about identification conditions in SOO settings.

We want to choose a **conditioning set** (X) that **blocks all back-door paths** between D and Y :

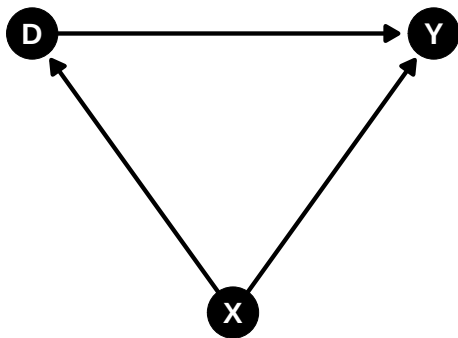
- Back-door path: A path that starts with an arrow into D and ends with an arrow into Y , includes no descendants of D , and does not have a collider.
- Collider: A node with two arrows into it, e.g., $D \rightarrow U \leftarrow Y$. This blocks a path.

SOO: Identification (DAGs)



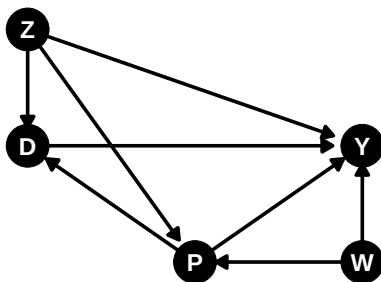
Is $\{X\}$ a sufficient conditioning set to identify the effect of D on Y ?

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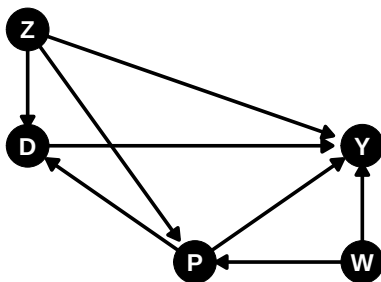
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SOO: Identification (DAGs)



Does $\{Z\}$ identify the effect of D on Y ?

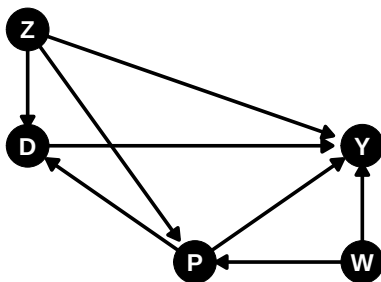
SOO: Identification (DAGs)



Does $\{Z\}$ identify the effect of D on Y ? No!

Does $\{P\}$ identify the effect of D on Y ?

SOO: Identification (DAGs)

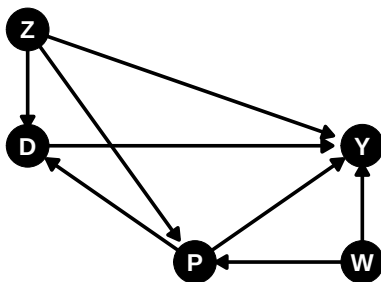


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SOO: Identification (DAGs)



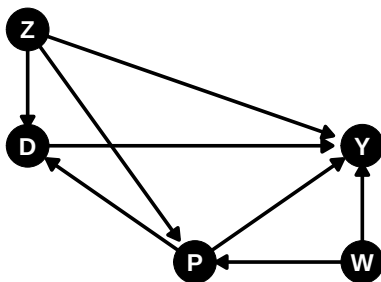
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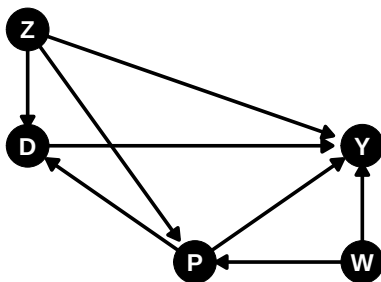
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SOO: Estimation

There are four main ways to estimate the effect of D on Y under SOO:

1. Stratification
2. Weighting
3. Matching
4. Regression

We will have to skip 1 and 2.

SOO: Estimation (Matching)

Consider the **propensity score**:

$$\pi(X_i) \equiv \Pr(D_i = 1 \mid X_i)$$

It turns out that if we can make our treatment and control groups equal in terms of $\pi(X_i)$, this is sufficient for identification.

(This is actually amazing)

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(This is actually amazing)

But we have to **estimate** $\pi(X_i)$.

Once we do that, we can **match** on $\hat{\pi}(X_i)$ – for every treated unit, find a control unit that looks ‘the same’ in terms of $\hat{\pi}(X_i)$.

SOO: Estimation (Matching)

Let's estimate the propensity score using a logistic regression model.

It turns out I know that Reactive is the key feature that drives treatment assignment (plus noise):

```
# Fit the logistic regression (you will do this again with Ryan!)
logit_fit <- glm(Medicine_Selection ~ Reactive,
                 data = students_analysis,
                 family = binomial(link = "logit"))

# Predict the propensity scores (note "type = "responses"")
students_analysis <- students_analysis %>%
  mutate(Propensity_Score = predict(logit_fit, type = "response"))
```

SOO: Estimation (Matching)

Let's do this 'properly' with MatchIt (many possible bells and whistles):

```
match_result <- MatchIt::matchit(  
  Medicine_Selection ~ Reactive,  
  data = students_analysis,  
  distance = students_analysis$Propensity_Score  
)  
  
match_result
```

A `matchit` object

- method: 1:1 nearest neighbor matching without replacement
- distance: User-defined - number of obs.: 1000 (original), 694 (matched)
- target estimand: ATT
- covariates: Reactive

Note: If you just specify 'method = "nearest"', it will estimate the propensity score itself and use that!

SOO: Estimation (Matching)

And now estimate the ATT:

```
matched_data <- match.data(match_result)

t.test(Headache ~ Medicine_Selection, data = matched_data)
```

Welch Two Sample t-test

data: Headache by Medicine_Selection

t = -0.73222, df = 691.98, p-value = 0.4643

alternative hypothesis: true difference in means between group 0 and group 1 is

95 percent confidence interval:

-0.7174173 0.3276669

sample estimates:

mean in group 0 mean in group 1

7.780188 7.975063

SOO: Estimation (Regression)

Finally, let's come back to our old friend, regression:

We can use regression to estimate the effect of D on Y under SOO, but we have to include X as a control:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 D_i + \hat{\beta}_2 X_i$$

Note that the interpretation of $\hat{\beta}_1$ has not changed!

Note also that $\hat{\beta}_2$ is *not* a causal effect – it is a nuisance parameter.

SOO: Estimation (Regression)

Let's do this in R:

```
lm_robust(Headache ~ Medicine_Selection + Reactive,  
          data = students_analysis) %>%  
  summary()
```

Call:

```
lm_robust(formula = Headache ~ Medicine_Selection + Reactive,  
          data = students_analysis)
```

Standard error type: HC2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	CI Lower	CI Upper	DF
(Intercept)	5.3240	0.1548	34.3996	1.352e-171	5.0203	5.6277	997
Medicine_Selection	0.1338	0.2497	0.5356	5.923e-01	-0.3563	0.6239	997
Reactive	2.5920	0.2266	11.4410	1.446e-28	2.1474	3.0366	997

Multiple R-squared: 0.1087 , Adjusted R-squared: 0.1069

F-statistic: 86.48 on 2 and 997 DF, p-value: < 2.2e-16

SOO: Estimation (Agnostic Regression)

One well-regarded large-sample linear regression estimator (Lin, 2013):

$$Y_i = \hat{\beta}_0 + D_i \hat{\beta}_{1\text{int}} + (X_i - \bar{X}) \hat{\beta}_2 + D_i (X_i - \bar{X}) \hat{\beta}_3$$

Where X_i are our covariates and $\bar{X}$ is the sample mean of X_i

Read: De-mean X , and interact it with the D .

You can do this in an experiment too – you will potentially gain some efficiency advantages.

SOO: Estimation (Agnostic Regression)

The `{estimatr}` package gives us a canned function for this:

```
lm_lin(Headache ~ Medicine_Selection, covariates = ~ Reactive,  
       data = students_analysis) %>%  
  summary()
```

Call:

```
lm_lin(formula = Headache ~ Medicine_Selection, covariates = ~Reactive,  
       data = students_analysis)
```

Standard error type: HC2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	CI Lower
(Intercept)	7.26304	0.1372	52.93796	9.100e-292	6.9938
Medicine_Selection	0.14034	0.2772	0.50631	6.128e-01	-0.4036
Reactive_c	2.59392	0.2361	10.98517	1.407e-26	2.1306
Medicine_Selection:Reactive_c	-0.03239	0.8262	-0.03921	9.687e-01	-1.6536

	CI Upper	DF
(Intercept)	7.5323	996
Medicine_Selection	0.6843	996
Reactive_c	3.0573	996
Medicine_Selection:Reactive_c	1.5888	996

Multiple R-squared: 0.1087 , Adjusted R-squared: 0.106

F-statistic: 57.54 on 3 and 996 DF, p-value: < 2.2e-16

Wrapping Up

Today we covered:

1. Fundamental building blocks for causal inference
2. How to 'solve' the FPCI with experiments
3. How to 'solve' the FPCI with conditioning

Remember: Both approaches are *assumption* driven!

Tomorrow:

1. Instrumental variables
2. Regression discontinuity
3. Difference-in-differences